DIVYANSH RAJAT

B.Tech CSE Student | AI/ML & Backend Developer | Smart India Hackathon 2026 Finalist

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SUMMARY

Final-year Computer Science undergraduate who builds end-to-end **AI/ML and backend systems** in Python (PyTorch, FastAPI, Flask, MongoDB). Lead architect on two **SIH 2026 finalist** projects, creator of a multimodal deepfake-detection platform, and author of **two published/presented research papers** in NLP and Responsible AI. Seeking an internship to contribute to AI/ML, backend and scalable software teams.

EDUCATION

B.Tech, Computer Science & Engineering — BBSBEC, I.K. Gujral Punjab Technical University | **CGPA: 7.98 / 10** *Jul 2023 – May 2027*

Class XII (CBSE, Non-Medical) – 59.9% (2017) | **Class X** (CBSE) – 75% (2015)

PROJECTS

CheckReality AI — Multimodal Deepfake & Misinformation Detection (Capstone) | GitHub *BBSBEC Major Project*

- Built a full-stack **FastAPI + PyTorch** platform detecting AI-generated text, face-swap images, deepfake video and cloned audio using RoBERTa, EfficientNet-B3 (+ BiLSTM for video) and a Mel-spectrogram CNN; models reach **99%+** accuracy on benchmark datasets.
- Designed a cross-modal **fusion engine** (weighted attention, Platt-scaling calibration) producing one risk score, with **explainable AI** (Grad-CAM heatmaps, SHAP token attribution) and auto-generated PDF audit reports; JWT auth, admin dashboard, Dockerised with MongoDB.

Tech: Python, PyTorch, FastAPI, Transformers, OpenCV, MongoDB, Docker, JavaScript

NyayaVault — Court-Ready Offline Evidence Vault (SIH 2026 Finalist, SIH26190) | GitHub *Team S.W.O.R.D.*

- Architected a **100% offline, air-gapped** evidence-management system for the Indian judicial system, running quantized NLP (LaMini-Flan-T5) **in the browser via WebGPU/WASM** with OCR and summarization and zero cloud telemetry.
- Implemented an append-only **SHA-256 hash-chain ledger** for tamper-evident logs, Section 63 BNSS-compliant reports, and 5-tier role/attribute-based access control.

Tech: Next.js, TypeScript, Prisma, Transformers.js, Tesseract.js, WebGPU/WASM

ULPF — Universal Log Pre-processing Framework (SIH 2026 Finalist, NTRO, SIH26156) | GitHub *Lead Architect*

- Led design of an air-gapped, lossless middleware that normalises heterogeneous raw logs (Syslog, NetFlow, etc.) into **OCSF/ECS** schemas with a sub-millisecond (**<0.8 ms**) parsing hot path.
- Split the system into an offline AI control plane (Qwen2.5-Coder generates parsing rules as ASTs) and a zero-AI-latency data plane; streamed events via **Redis Streams** into **OpenSearch** with a write-once, SHA-256-verified raw vault.

Tech: Python, FastAPI, Redis Streams, OpenSearch, PostgreSQL, Qwen2.5-Coder

AI-Based Plagiarism Detection System | GitHub *Jan 2026 – May 2026*

- Built a **Flask** web app that detects paraphrased plagiarism using **Sentence-BERT (MiniLM) embeddings and cosine similarity**, plus an AI-generated-text detection module based on perplexity and stylometry.
- Created a pipeline for PDF/DOCX/TXT extraction (with OCR fallback), role-based auth (Student/Teacher/Admin) on **MongoDB**, and structured similarity reports.

Tech: Python, Flask, NLTK, Sentence-Transformers, MongoDB, HTML/CSS

RESEARCH & PUBLICATIONS

Ethical and Responsible AI: A Comprehensive Review of Principles, Methods, and Tools — Published in Int. Journal of Information Security Engineering (IJISE), Vol. 4, Issue 1 (2026); presented at **ICIAI 2025** (IEEE Chandigarh Section, DST-supported). Covers fairness, SHAP/LIME, federated learning, differential privacy.

Natural Language Processing in Software Applications: A Comprehensive Review — Published in ISBN-edited book (978-81-966131-4-3); presented at the **37th National Convention of Computer Engineers & AIML-2024**. Covers BERT, GPT, sentiment analysis, chatbots, machine translation.

TECHNICAL SKILLS

Languages: Python, C++, SQL, TypeScript, JavaScript, Bash

AI / ML: Deep Learning (CNNs, LSTMs, Transformers), NLP, Computer Vision, PyTorch, Scikit-learn, NumPy, Pandas, OpenCV, Hugging Face, Sentence-Transformers, Explainable AI (SHAP, Grad-CAM)

Backend & Databases: FastAPI, Flask, REST APIs, JWT Auth, Redis, Celery, MongoDB, PostgreSQL, MySQL, Prisma

Tools & Cloud: Git/GitHub, Docker, Linux, AWS, CI/CD, Postman, Jupyter/Colab, React/Next.js (basic)

Security: Secure Coding, OWASP, API Security, Cryptography (RSA, SHA-256)

CS Core: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP, Software Engineering, Design Patterns

ACHIEVEMENTS & LEADERSHIP

- **Smart India Hackathon 2026 Finalist** (Team S.W.O.R.D.) — lead architect for two national problem statements (SIH26156 NTRO, SIH26190).
- **3rd Position**, Saviskar 2025 Tech Event, CGC University, Mohali (Oct 2025). **Participant:** Quantum 3.0 Vibe Coding Hackathon, Hack-N-Win 3.0 Hackathon, Junk2Genius 2025 National Ideathon.
- **Lead Coordinator**, College Tech Fest (Apr 2026) and **Organizer**, Alumni Meet (Nov 2025) — event planning, team coordination, registration and technical execution.

CERTIFICATIONS

Machine Learning & Python Workshop, NIELIT Chandigarh (Mar 2025) | **Cyber Security Certification**, NIELIT Delhi (Jun–Jul 2025)